

AI-SOC Assistant:

AI-Powered Security Alert Analysis and Incident Reporting System



Personal Cybersecurity & SOC Automation Project

Final Version: v4.8

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Technologies:

- Python
- Wazuh
- Google Gemini API
- REST APIs
- Threat Intelligence APIs
- JSON
- Git & GitHub
- Visual Studio Code
- Windows/Linux
- Python Virtual Environment (venv)

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TABLE OF CONTENTS

1. PROJECT OVERVIEW	4
2. PROJECT OBJECTIVES	5
3. PROBLEM STATEMENT	6
4. PROJECT SCOPE	7
5. TECHNOLOGIES USED	8
6. SYSTEM ARCHITECTURE	9
7. SYSTEM COMPONENTS	11
7.1 Parser	11
7.2 Alert Formatter	11
7.3 Threat Intelligence Manager	11
7.4 Risk Engine	11
7.5 Incident Classifier	11
7.6 Evidence Assessment	11
7.7 Context Builder	12
7.8 AI SOC Analysis	12
7.9 Report Generator	12
7.10 Main Pipeline	12
8. SYSTEM WORKFLOW	13
Step 1 — Alert Input	13
Step 2 — Alert Parsing	13
Step 3 — Alert Formatting	13
Step 4 — Threat Intelligence	13
Step 5 — Risk Assessment	13
Step 6 — Incident Classification	13
Step 7 — Evidence Assessment	13
Step 8 — Context Construction	14
Step 9 — AI SOC Analysis	14
Step 10 — Incident Report Generation	14

Step 11 — Report Storage	14
9. IMPLEMENTATION DETAILS.....	16
9.2 Main Application.....	17
9.3 Alert Parsing	17
9.4 Threat Intelligence.....	17
9.5 Deterministic Risk Engine	18
9.6 Evidence-Based Analysis	18
9.7 AI Analysis.....	18
9.8 Report Generation	18
10. TESTING AND VALIDATION	19
10.1 Alert Input Testing	19
10.2 Alert Formatting Testing.....	19
10.3 Threat Intelligence Testing	19
10.4 Risk Assessment Testing.....	19
10.5 Incident Classification Testing	19
10.6 Evidence Assessment Testing	19
10.7 AI SOC Analysis Testing	20
10.8 Report Generation Testing	20
10.9 End-to-End Validation	20
11. RESULTS AND OUTPUT	21
11.1 Sample Alert.....	21
12. LIMITATIONS AND FUTURE IMPROVEMENTS	31
12.1 Limitations	31
12.2 Future Improvements.....	31
13. CONCLUSION	32
14. REFERENCES	33
14.1 Cybersecurity and SOC References	33
14.2 Python References.....	33
14.3 AI / LLM Reference	33
15. APPENDICES	34

1. PROJECT OVERVIEW

AI-SOC Assistant is an independent personal cybersecurity project designed to demonstrate an evidence-based approach to Security Operations Center (SOC) incident analysis and reporting. The system processes security alerts, enriches the available alert information with cybersecurity knowledge, performs deterministic risk assessment, classifies the detected incident, evaluates the available evidence, and uses AI-assisted analysis to produce a structured SOC incident report.

The project focuses on the practical workflow followed during security alert investigation. Rather than relying solely on an AI model to determine the severity or conclusion of an incident, the system separates deterministic security processing from AI-assisted reasoning. Wazuh provides the security alert, while the application processes the alert through multiple analysis stages before generating the final incident report.

The finalized **v4.8 architecture** follows the pipeline:

Wazuh Alert → Parser → Alert Formatter → Threat Intelligence → Risk Engine → Incident Classification → Evidence Assessment → AI SOC Analysis → Incident Report

The system uses MITRE ATT&CK techniques associated with security alerts to provide additional cybersecurity context. Threat intelligence is applied where appropriate, including handling private internal IP addresses separately from publicly accessible IP addresses. A deterministic risk engine calculates the incident risk score using predefined factors such as Wazuh rule severity, IP classification, threat intelligence reputation, and associated MITRE techniques.

The evidence assessment stage separates confirmed information from suspicious observations and unconfirmed possibilities. This approach helps prevent the AI analysis from presenting assumptions as confirmed security events. The AI SOC analysis therefore works from the structured context produced by the preceding stages and provides an investigation-oriented interpretation, recommended actions, and a final SOC analyst conclusion.

The final output is a Markdown-based incident report containing the incident context, risk assessment, classification, evidence assessment, AI SOC analysis, and report information. Version **v4.8** represents the finalized and frozen implementation of the project.

2. PROJECT OBJECTIVES

The main objective of the **AI-SOC Assistant** is to develop a practical, evidence-based system that assists with the analysis of security alerts and the generation of structured SOC incident reports.

The specific objectives of the project are:

1. **Process security alerts** by loading and parsing Wazuh-formatted alert data.
2. **Format alert information** into a clear and human-readable representation containing rule information, agent information, source IP, timestamp, MITRE IDs, and associated cybersecurity knowledge.
3. **Integrate MITRE ATT&CK knowledge** to provide contextual information about techniques associated with security alerts.
4. **Perform threat intelligence analysis** for applicable source IP addresses and distinguish between private internal addresses and publicly accessible addresses.
5. **Perform deterministic risk assessment** using predefined security factors such as Wazuh rule severity, IP classification, threat intelligence reputation, and MITRE techniques.
6. **Classify security incidents** according to their incident type, attack category, primary technique, and protocol.
7. **Assess available evidence** by separating confirmed evidence, suspicious observations, and unconfirmed possibilities.
8. **Use AI-assisted SOC analysis** to interpret the structured incident context while maintaining a clear distinction between confirmed facts and assumptions.
9. **Generate structured incident reports** containing the incident context, analysis, severity, and investigation recommendations.
10. **Maintain an evidence-based analysis workflow** that avoids claiming successful compromise, lateral movement, credential compromise, or other security events without supporting evidence.

3. PROBLEM STATEMENT

Security Operations Centers receive a large number of security alerts from monitoring and detection systems such as Wazuh. These alerts often contain technical information including rule identifiers, severity levels, source IP addresses, timestamps, and MITRE ATT&CK mappings. Manually reviewing this information and converting it into a structured incident assessment can be time-consuming and may lead to inconsistent analysis.

A further challenge is that a security alert does not necessarily prove that an attack or compromise was successful. For example, repeated SSH authentication failures may indicate password-guessing activity, but the alert alone does not confirm successful authentication, credential compromise, lateral movement, or host compromise. Therefore, an effective SOC analysis process must distinguish between confirmed evidence, suspicious observations, and unconfirmed possibilities.

The **AI-SOC Assistant** addresses this problem by providing a structured processing pipeline for security alerts. The system combines deterministic processing, MITRE ATT&CK contextualization, applicable threat intelligence, risk assessment, incident classification, evidence assessment, and AI-assisted analysis. The deterministic components establish the factual and risk-related context, while the AI component analyzes that structured context without replacing the underlying evidence.

The system ultimately converts a raw security alert into a structured SOC incident report containing the relevant alert information, threat intelligence status, risk assessment, incident classification, evidence assessment, AI-assisted analysis, investigation steps, remediation recommendations, and a final SOC conclusion.

4. PROJECT SCOPE

The scope of the **AI-SOC Assistant** is focused on the automated processing and analysis of security alerts within a SOC-oriented workflow. The finalized v4.8 implementation is designed to demonstrate how alert information can be transformed into structured security analysis and an incident report.

The system covers the following areas:

1. **Security Alert Processing** — Loading and parsing Wazuh alert data provided to the system.
2. **Alert Formatting** — Converting raw alert information into a readable security alert containing relevant rule, agent, source IP, timestamp, and MITRE information.
3. **MITRE ATT&CK Context** — Providing cybersecurity knowledge for MITRE techniques associated with the alert.
4. **Threat Intelligence** — Performing applicable IP reputation analysis while identifying private internal IP addresses for which public threat-intelligence lookup is not applicable.
5. **Risk Assessment** — Calculating a deterministic risk score and assigning a corresponding risk level, incident status, and confidence level.
6. **Incident Classification** — Categorizing alerts according to incident type, attack category, primary technique, and protocol.
7. **Evidence Assessment** — Separating confirmed evidence, suspicious observations, and unconfirmed conditions.
8. **AI-Assisted Analysis** — Providing SOC-oriented interpretation and recommendations based on the structured incident context.
9. **Incident Reporting** — Generating and saving a structured Markdown incident report.

The project is intentionally limited to the finalized **v4.8 architecture**. It does not attempt to replace a complete enterprise SOC, perform autonomous incident response, or confirm compromise without supporting evidence. The AI component assists with analysis and reporting, while deterministic processing remains responsible for important security assessments.

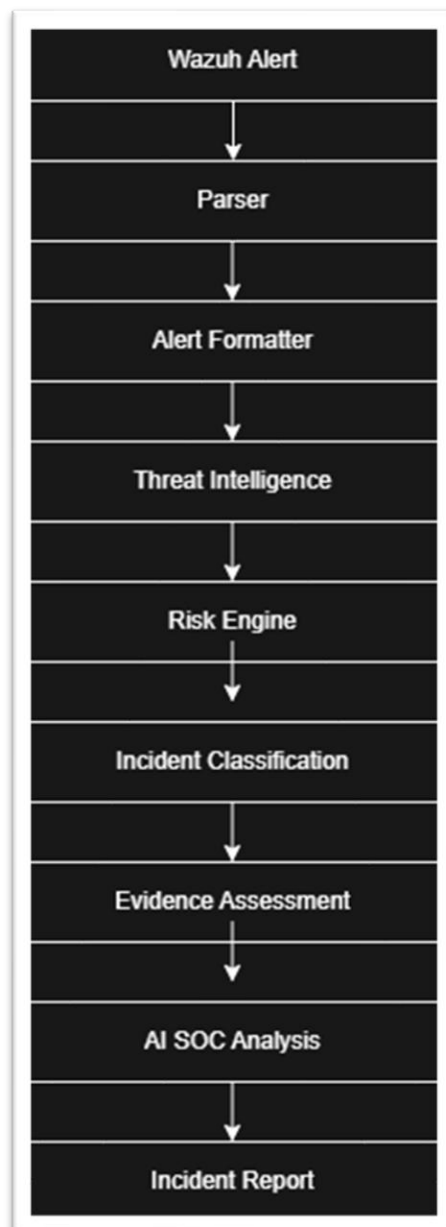
5. TECHNOLOGIES USED

The AI-SOC Assistant was developed using a combination of programming technologies, cybersecurity tools, threat-intelligence services, and AI technologies.

Technology / Tool	Purpose
Python	Core programming language used to develop the SOC analysis pipeline.
Wazuh	Source of security alerts used as input for the analysis workflow.
MITRE ATT&CK	Provides contextual information about adversary tactics and techniques associated with alerts.
AbuseIPDB	Provides IP reputation and abuse-confidence information for applicable public IP addresses.
OpenRouter LLM	Provides AI-assisted SOC analysis based on the structured incident context.
JSON	Used for representing and storing sample security alert data.
Markdown	Used as the format for generated SOC incident reports.
Git / GitHub	Used for version control and maintaining project versions, including the finalized v4.8 state.
Visual Studio Code	Development environment used to write, organize, test, and manage the project.
Windows / PowerShell	Development and execution environment used during project implementation and testing.

6. SYSTEM ARCHITECTURE

The finalized AI-SOC Assistant follows a sequential SOC analysis pipeline in which each stage processes the output of the previous stage. The architecture separates alert processing, deterministic security assessment, evidence evaluation, and AI-assisted analysis to maintain an evidence-based workflow.



1. Wazuh Alert

Provides the initial security alert containing information such as the rule ID, rule level, description, agent, source IP, timestamp, and MITRE ATT&CK information.

2. Parser

Loads the alert data and converts the JSON alert into a Python data structure that can be processed by the remaining components.

3. Alert Formatter

Converts the alert information into a human-readable format and adds locally maintained MITRE ATT&CK cybersecurity knowledge.

4. Threat Intelligence

Analyzes applicable indicators, particularly source IP addresses. Private internal IP addresses are identified separately, while applicable public IP addresses can be checked through AbuseIPDB.

5. Risk Engine

Performs deterministic risk scoring using predefined factors such as Wazuh rule severity, IP classification, AbuseIPDB reputation, and associated MITRE techniques.

6. Incident Classification

Determines the incident type, attack category, primary MITRE technique, and protocol based on the available alert information.

7. Evidence Assessment

Separates information into confirmed evidence, suspicious observations, and unconfirmed possibilities. This prevents assumptions from being treated as confirmed security events.

8. AI SOC Analysis

Receives the structured incident context and performs AI-assisted SOC analysis. The AI provides interpretation, investigation steps, remediation recommendations, and a final conclusion while following the evidence-based requirements.

9. Incident Report

Combines the incident context and AI analysis into a structured Markdown report and saves it in the project's `reports` directory.

7. SYSTEM COMPONENTS

7.1 Parser

The parser is responsible for loading the Wazuh alert from a JSON file. It reads the alert using Python's JSON processing functionality and converts the stored alert into a Python dictionary. This provides the structured input required by the remaining stages of the SOC analysis pipeline.

7.2 Alert Formatter

The alert formatter converts the structured Wazuh alert into a human-readable security alert. It extracts rule information, agent information, source IP, timestamp, MITRE IDs, and MITRE tactics. It also incorporates locally maintained cybersecurity knowledge associated with the MITRE techniques present in the alert.

7.3 Threat Intelligence Manager

The Threat Intelligence Manager determines whether threat-intelligence analysis is applicable to the source indicator. It identifies the indicator type and handles private internal IP addresses separately because they are not applicable to public IP reputation platforms. For applicable public IP addresses, the system can use AbuseIPDB to obtain reputation information and an abuse-confidence score.

7.4 Risk Engine

The Risk Engine is a deterministic component responsible for calculating the security risk score. It evaluates predefined factors including Wazuh rule severity, source IP classification, AbuseIPDB reputation, and associated MITRE techniques. The resulting score is mapped to a risk level and accompanied by an incident status and confidence value.

7.5 Incident Classifier

The Incident Classifier categorizes the detected security event. It determines the incident type, attack category, primary MITRE technique, and protocol by evaluating information contained in the Wazuh alert.

7.6 Evidence Assessment

The Evidence Assessment component organizes the available information into three categories: confirmed evidence, suspicious observations, and unconfirmed conditions. This separation is an important part of the project's evidence-based approach because the presence of a security alert or an associated MITRE technique does not by itself prove that a successful attack or compromise occurred.

7.7 Context Builder

The Context Builder combines the outputs of the alert formatter, threat intelligence, risk engine, incident classifier, and evidence assessment into a structured incident context. This context becomes the input provided to the AI SOC analysis component.

7.8 AI SOC Analysis

The AI SOC Analysis component uses an OpenRouter-based large language model to analyze the structured incident context. It produces an executive summary, attack assessment, investigation steps, remediation recommendations, and a SOC analyst conclusion. The analysis is constrained by the supplied evidence and is instructed not to present unconfirmed events as established facts.

7.9 Report Generator

The Report Generator creates the final SOC incident report in Markdown format. The report combines the incident context and AI SOC analysis and stores the resulting report inside the project's reports directory.

7.10 Main Pipeline

The main application coordinates the complete processing sequence. It loads the alert, executes each analysis component in the defined order, displays the results in the console, generates the final AI-assisted analysis, and saves the resulting incident report.

8. SYSTEM WORKFLOW

The AI-SOC Assistant processes a security alert through a sequence of controlled analysis stages. Each stage contributes specific information to the next stage, resulting in a structured SOC incident report.

Step 1 — Alert Input

The system receives a Wazuh security alert containing information such as the rule ID, rule level, description, agent, source IP, timestamp, and MITRE ATT&CK information.

Step 2 — Alert Parsing

The parser loads the JSON alert and converts it into structured data that can be accessed by the application.

Step 3 — Alert Formatting

The alert formatter extracts the important alert fields and presents them in a human-readable format. Associated MITRE techniques are enriched with locally maintained cybersecurity knowledge.

Step 4 — Threat Intelligence

The source indicator is evaluated to determine whether threat-intelligence lookup is applicable. Private internal IP addresses are identified as internal indicators and are not submitted to public IP reputation platforms. Public IP addresses can be checked using AbuseIPDB.

Step 5 — Risk Assessment

The Risk Engine calculates a deterministic risk score using predefined security factors. The resulting score is converted into a risk level such as **LOW, MEDIUM, HIGH, or CRITICAL**.

Step 6 — Incident Classification

The alert is classified according to its incident type, attack category, primary technique, and protocol.

Step 7 — Evidence Assessment

The system separates the available information into **Confirmed Evidence**, **Suspicious Observations**, and **Unconfirmed** conditions. This ensures that an alert is not automatically treated as proof of compromise.

Step 8 — Context Construction

The Context Builder combines the alert, threat-intelligence results, risk assessment, incident classification, and evidence assessment into a structured context for AI analysis.

Step 9 — AI SOC Analysis

The structured context is provided to the AI analysis component. The AI generates a professional SOC assessment while following the evidence-based requirements defined by the system.

Step 10 — Incident Report Generation

The system combines the structured incident context and AI analysis into a Markdown incident report.

Step 11 — Report Storage

The generated report is saved inside the project's `reports` directory for later review and documentation.

Workflow Diagram Underneath:



9. IMPLEMENTATION DETAILS

9.1 Project Structure

AI-SOC-Assistant/

```
|
|
|─ app/
|   |─ abuseipdb.py
|   |─ ai.py
|   |─ alert_formatter.py
|   |─ config.py
|   |─ context_builder.py
|   |─ detector.py
|   |─ evidence.py
|   |─ incident_classifier.py
|   |─ ip_utils.py
|   |─ main.py
|   |─ mitre.py
|   |─ output.py
|   |─ parser.py
|   |─ prompts.py
|   |─ report.py
|   |─ risk_engine.py
|   |─ threat_rating.py
|   |─ ti_formatter.py
|   └─ ti_manager.py
```



```
|
|— docs/
|— reports/
|— screenshots/
|— sample_alerts/
|   └─ ssh_bruteforce.json
|
|— .env
|— .gitignore
|— LICENSE
|— README.md
└─ requirements.txt
```

9.2 Main Application

The `main.py` module coordinates the complete SOC analysis workflow. It loads the alert, formats the alert information, obtains threat-intelligence information, calculates the deterministic risk assessment, classifies the incident, evaluates evidence, builds the AI context, performs AI-assisted analysis, generates the incident report, and saves the report.

9.3 Alert Parsing

The `parser.py` module provides the alert-loading functionality. It reads a JSON-formatted Wazuh alert using Python's JSON library and returns the resulting data structure for further processing.

9.4 Threat Intelligence

The threat-intelligence functionality is divided between the AbuseIPDB integration, IP classification, threat-intelligence management, and result formatting. The system first determines whether the source IP is private or public. Private internal addresses are identified as internal indicators and public threat-intelligence lookup is skipped. Applicable public IP addresses can be queried through AbuseIPDB.

9.5 Deterministic Risk Engine

The `risk_engine.py` module calculates the incident risk using predefined rules. The score is influenced by Wazuh rule severity, IP classification, AbuseIPDB reputation, and selected MITRE techniques. The final score is limited to a maximum of 100 and mapped to a corresponding risk level.

9.6 Evidence-Based Analysis

The `evidence.py` and `context_builder.py` modules support the evidence-based analysis approach. Confirmed facts are separated from suspicious observations and unconfirmed conditions before the information is provided to the AI analysis component.

9.7 AI Analysis

The `ai.py` module provides the AI-assisted SOC analysis functionality. The AI receives the structured incident context rather than an unstructured raw alert. The system prompt establishes requirements for evidence-based reasoning and prevents unsupported claims from being presented as confirmed facts.

9.8 Report Generation

The `report.py` module generates the final Markdown incident report. It includes an incident identifier, generation time, severity, incident context, AI SOC analysis, and report information. Generated reports are stored in the `reports` directory.

10. TESTING AND VALIDATION

Testing was performed to verify that the finalized AI-SOC Assistant pipeline operates correctly from alert input through incident report generation. The testing focused on validating individual processing stages as well as the complete end-to-end workflow.

10.1 Alert Input Testing

A sample SSH brute-force Wazuh alert was used to test the system. The alert was represented in JSON format and contained a rule ID, rule severity, description, agent information, source IP, timestamp, and MITRE ATT&CK information.

10.2 Alert Formatting Testing

The alert formatter was tested to verify that the relevant Wazuh fields were correctly extracted and displayed. MITRE technique IDs were also checked to ensure that the corresponding locally maintained cybersecurity information was included in the formatted alert.

10.3 Threat Intelligence Testing

Threat-intelligence processing was tested using the source IP `192.168.192.140`. Because the address is a private internal IP address, the system correctly classified it as an internal address and skipped the public AbuseIPDB lookup.

10.4 Risk Assessment Testing

The deterministic risk engine was tested using the alert's rule severity, source IP classification, AbuseIPDB score, and MITRE techniques. For the tested alert, the system produced a risk score of **55/100**, corresponding to a **HIGH** risk level. The incident status was reported as **SUSPICIOUS** with **MEDIUM** confidence.

10.5 Incident Classification Testing

The tested alert was classified as an **SSH Brute Force** incident under the **Credential Attack** category. The primary technique was identified as **T1110.001 – Password Guessing**, with **SSH** identified as the protocol.

10.6 Evidence Assessment Testing

The evidence assessment successfully separated the available information into confirmed evidence, suspicious observations, and unconfirmed conditions. The system explicitly

identified that successful authentication, credential compromise, lateral movement, command execution, and host compromise had not been confirmed.

10.7 AI SOC Analysis Testing

The AI analysis was tested using the structured context generated by the previous processing stages. The resulting analysis provided an executive summary, attack assessment, investigation steps, remediation recommendations, and a SOC analyst conclusion. The analysis maintained the distinction between confirmed evidence and possible interpretations.

10.8 Report Generation Testing

The report-generation stage was tested to verify that the final incident report was successfully created in Markdown format and stored inside the `reports` directory. The generated report contained the incident context and AI SOC analysis.

10.9 End-to-End Validation

Test Area	Expected Result	Result
Alert Loading	JSON alert loaded successfully	PASS
Alert Formatting	Wazuh and MITRE information displayed	PASS
Private IP Detection	Internal IP identified correctly	PASS
Threat Intelligence	Public lookup skipped for private IP	PASS
Risk Assessment	Deterministic score generated	PASS
Incident Classification	SSH brute-force classification generated	PASS
Evidence Assessment	Evidence separated correctly	PASS
AI Analysis	SOC analysis generated	PASS
Report Generation	Markdown report created and saved	PASS

11. RESULTS AND OUTPUT

11.1 Sample Alert

The system was tested using a sample SSH brute-force Wazuh alert containing multiple SSH authentication failures from the internal source IP 192.168.192.140.

```
===== WAZUH ALERT =====  
  
Rule ID: 5763  
Rule Level: 10  
Description: Multiple SSH authentication failures  
  
Agent: ubuntu-agent  
  
Source IP: 192.168.192.140  
  
Timestamp: 2026-08-04T19:30:15Z  
  
MITRE IDs: T1110.001, T1021.004  
  
MITRE Tactics: Credential Access, Lateral Movement  
  
Cybersecurity Knowledge  
  
Technique: Password Guessing  
Tactic: Credential Access  
Description: Attackers attempt to guess passwords for valid user accounts.  
Mitigation: Use MFA, account lockout policies, and strong passwords.  
  
Technique: SSH  
Tactic: Lateral Movement  
Description: Attackers use SSH to move laterally or remotely access systems.  
Mitigation: Restrict SSH access, use key authentication, and monitor login attempts.
```

Figure 1: Sample Wazuh alert processed by the AI-SOC Assistant

11.2 Risk Assessment Result

```
===== RISK ASSESSMENT =====

Risk Score : 55/100
Risk Level : HIGH
Incident Status : SUSPICIOUS
Confidence : MEDIUM

Risk Factors:
- Wazuh Rule Level 10+: +30
- Private Internal IP: +5
- AbuseIPDB Score 0%: +0
- MITRE T1110.001 Password Guessing: +10
- MITRE T1021.004 SSH: +10
```

Figure 2: Deterministic risk assessment output

The deterministic Risk Engine assigned the tested alert a score of **55/100**, resulting in a **HIGH** risk classification. The incident remained **SUSPICIOUS**, while the confidence level was **MEDIUM**.

11.3 Incident Classification Result

The system classified the alert as an **SSH Brute Force** incident under the **Credential Attack** category. The primary MITRE ATT&CK technique was identified as **T1110.001 – Password Guessing**, with SSH identified as the protocol.

```
===== INCIDENT CLASSIFICATION =====  
  
Incident Type : SSH Brute Force  
Attack Category : Credential Attack  
Primary Technique : T1110.001 - Password Guessing  
Protocol : SSH
```

Figure 3: Incident classification output

11.4 Evidence Assessment Result

The Evidence Assessment stage separated the available information into confirmed evidence, suspicious observations, and unconfirmed conditions. This demonstrated the system's evidence-based approach and prevented unverified compromise claims from being treated as confirmed findings.

```
===== EVIDENCE ASSESSMENT =====  
  
Confirmed Evidence:  
- Wazuh rule 5763 generated the alert.  
- Rule description: Multiple SSH authentication failures.  
- Rule level: 10.  
- Source IP observed: 192.168.192.140.  
- MITRE techniques associated with the alert: T1110.001, T1021.004.  
  
Suspicious Observations:  
- The observed authentication failures may indicate password-guessing activity.  
- The activity requires investigation to determine whether the source is authorized or potentially compromised.  
  
Unconfirmed:  
- Successful authentication has not been confirmed.  
- Credential compromise has not been confirmed.  
- Lateral movement has not been confirmed.  
- Command execution has not been confirmed.  
- Host compromise has not been confirmed.
```

Figure 4: Evidence assessment output

11.5 AI SOC Analysis Result

The AI SOC Analysis produced an executive summary, attack assessment, investigation steps, remediation recommendations, and a final SOC analyst conclusion. The generated analysis maintained the distinction between confirmed observations and unconfirmed possibilities.

```
===== AI SOC ANALYSIS =====

## Executive Summary
A Wazuh rule (5763) flagged multiple failed SSH logins originating from the internal address **192.168.192.140**. The alert is classified as a credential-guessing attempt (MITRE T1110.001) and potentially a lateral-movement probe (T1021.004). No evidence of successful authentication, credential compromise, or subsequent activity has been observed. The incident is therefore **suspicious**;
```

Category	Observation
Confirmed behavior	Wazuh generated a Level 10 alert for multiple SSH authentication failures. Source IP is 192.168.192.140 (private/internal). MITRE techniques T1110.001 and T1021.004 are associated with the alert.
Reasonable interpretation	The pattern of repeated failures suggests a password-guessing or brute-force attempt. The internal IP could belong to an authorized administrator, a scheduled scan, or a compromised workstation.
Important uncertainty	No successful login or subsequent activity has been recorded. The legitimacy of the source IP and the intent behind the failures remain unknown.

```
## Investigation Steps

1. **Review SSH authentication logs** on the target host for any successful logins from 192.168.192.140 or related timestamps.
2. **Cross-reference the source IP** against the organization's asset inventory and known administrative devices.
3. **Check for scheduled scans or automated tools** that might target SSH on internal hosts, and verify their credentials.
4. **Inspect the originating host** (192.168.192.140) for signs of compromise: unusual processes, outbound connections, or recent configuration changes.
5. **Validate account lockout and MFA settings** for the accounts targeted by the failures to assess potential impact.

## Recommended Remediation

- **If the source is unauthorized,** consider blocking or rate-limiting SSH access from 192.168.192.140 pending confirmation.
- **Enforce MFA and strong password policies** for all SSH accounts to mitigate future brute-force attempts.
- **Implement account lockout thresholds** to automatically suspend accounts after a defined number of failed attempts.
- **Restrict SSH access** to known administrative IP ranges and enforce key-based authentication where possible.
- **Monitor SSH logs continuously** for repeated failure patterns and integrate alerts with a SIEM for real-time correlation.

## SOC Analyst Conclusion

**Classification:** Suspicious
**Rationale:** The alert confirms repeated SSH authentication failures from an internal IP, but there is no evidence of successful login, credential compromise, or lateral movement. The source could be legitimate or malicious; further evidence is required to confirm compromise.

**Evidence is insufficient to confirm compromise.**
```

Figure 5: AI-generated SOC analysis

The AI SOC Analysis produced an executive summary, attack assessment, investigation steps, remediation recommendations, and a final SOC analyst conclusion. The generated analysis maintained the distinction between confirmed observations and unconfirmed possibilities.

11.6 Generated Incident Report

After completing the analysis pipeline, the system generated a Markdown-based incident report and stored it in the `reports` directory. The report combines the structured incident context with the AI SOC analysis and provides a documented representation of the investigated alert.

```
reports > Incident_Report_20260817_222223.md > abc # Evidence Disclaimer
1  # AI SOC INCIDENT REPORT
2
3  **Incident ID:** IR-20260817-222223
4
5  **Generated:** 17 August 2026, 10:22:23 PM
6
7  **Severity:** HIGH
8
9  **Risk Score:** 55/100
10
11 **Incident Status:** SUSPICIOUS
12
13 **Confidence:** MEDIUM
14
15 ---
16
17 # Incident Context
18
19 ===== INCIDENT CONTEXT =====
20
21 ### WAZUH ALERT
22
23 Rule ID: 5763
24 Rule Level: 10
25 Description: Multiple SSH authentication failures
26
27 Agent: ubuntu-agent
28
29 Source IP: 192.168.192.140
30
```

Figure 6: Generated SOC incident report

```

31  Timestamp: 2026-08-04T19:30:15Z
32
33  MITRE IDs: T1110.001, T1021.004
34
35  MITRE Tactics: Credential Access, Lateral Movement
36
37  Cybersecurity Knowledge
38
39  Technique: Password Guessing
40  Tactic: Credential Access
41  Description: Attackers attempt to guess passwords for valid user accounts.
42  Mitigation: Use MFA, account lockout policies, and strong passwords.
43
44  Technique: SSH
45  Tactic: Lateral Movement
46  Description: Attackers use SSH to move laterally or remotely access systems.
47  Mitigation: Restrict SSH access, use key authentication, and monitor login attempts.
48
49  =====
50
51  ### THREAT INTELLIGENCE
52
53  ===== Threat Intelligence =====
54
55  IP Address : 192.168.192.140
56
57  IP Type : Private Internal Address

```

Figure 7: Generated SOC incident report

```
58
59  Threat Intelligence : Not Applicable
60
61  Reason :
62  Private IP addresses are not indexed by public
63  Threat Intelligence platforms.
64
65  Threat Rating : ● INTERNAL
66
67  Lookup Status : SKIPPED
68
69  =====
70
71  ### DETERMINISTIC RISK ASSESSMENT
72
73  Risk Score:
74  55/100
75
76  Risk Level:
77  HIGH
78
79  Incident Status:
80  SUSPICIOUS
81
82  Confidence:
83  MEDIUM
84
85  =====
```

Figure 8: Generated SOC incident report

```

113  ### SUSPICIOUS OBSERVATIONS
114
115  - The observed authentication failures may indicate password-guessing activity.
116  - The activity requires investigation to determine whether the source is authorized or potentially compromised.
117
118  =====
119
120  ### UNCONFIRMED
121
122  - Successful authentication has not been confirmed.
123  - Credential compromise has not been confirmed.
124  - Lateral movement has not been confirmed.
125  - Command execution has not been confirmed.
126  - Host compromise has not been confirmed.
127
128  =====
129
130  ### SOC ANALYSIS REQUIREMENTS
131
132  Analyze this incident as a professional SOC analyst.
133
134  IMPORTANT:
135
136  1. Clearly separate confirmed facts from reasonable interpretations
137     | and unconfirmed possibilities.
138
139  2. Do not claim compromise unless the supplied evidence confirms it.
140

```

```

87  ### INCIDENT CLASSIFICATION
88
89  Incident Type:
90  SSH Brute Force
91
92  Attack Category:
93  Credential Attack
94
95  Primary Technique:
96  T1110.001 - Password Guessing
97
98  Protocol:
99  SSH
100
101  =====
102
103  ### CONFIRMED EVIDENCE
104
105  - Wazuh rule 5763 generated the alert.
106  - Rule description: Multiple SSH authentication failures.
107  - Rule level: 10.
108  - Source IP observed: 192.168.192.140.
109  - MITRE techniques associated with the alert: T1110.001, T1021.004.
110
111  =====

```

Figure 9,10: Generated SOC incident report

```

140
141 3. Do not claim successful authentication unless successful
142 | authentication is explicitly present in the evidence.
143
144 4. Do not claim lateral movement unless there is direct evidence
145 | supporting lateral movement.
146
147 5. Do not treat the absence of threat intelligence as proof that
148 | an IP address is safe.
149
150 6. Consider threat intelligence when assessing the incident.
151
152 7. If threat intelligence conflicts with the observed Wazuh behavior,
153 | explicitly describe the conflict and consider both sources.
154
155 8. Respect the deterministic risk assessment and incident classification,
156 | but do not treat either one as proof of compromise.
157
158 9. Treat MITRE ATT&CK mappings as technique classifications,
159 | not proof that the technique was successfully executed.
160
161 10. Explain why the incident is suspicious, likely benign,
162 | confirmed malicious, or inconclusive based on the evidence.
163
164 11. Recommend investigation steps based only on available evidence.
165
166 12. Do not invent usernames, processes, malware, commands,
167 | successful logins, network activity, or other events.
168
169 13. For private/internal IP addresses, consider legitimate administrative
170 | activity, authorized security scanning, testing, or a potentially
171 | compromised internal host as possible explanations.
172
173 14. Do not overstate confidence.
174
175 15. If the available evidence is insufficient to establish compromise,
176 | explicitly state:
177 |
178 | "Evidence is insufficient to confirm compromise."
179
180 16. The final conclusion must be based on supplied evidence,
181 | not assumptions or external information.
182
183 =====

```

Figure 11,12: Generated SOC incident report

```

186
187 # 🛡️ AI SOC Analysis
188
189 ## Executive Summary
190 A Wazuh rule (5763) flagged multiple failed SSH logins originating from the internal address **192.168.192.140**. The alert is classified
as a credential-guessing attempt (MITRE T1110.001) and potentially a lateral-movement probe (T1021.004). No evidence of successful
authentication, credential compromise, or subsequent activity has been observed. The incident is therefore **suspicious**; further
investigation is required to determine whether the source is an authorized user, a legitimate security scan, or an attacker-controlled
host.
191
192 ## Attack Assessment
193
194 | Category | Observation |
195 |-----|-----|
196 | **Confirmed behavior** | • Wazuh generated a Level 10 alert for multiple SSH authentication failures.<br>• Source IP is 192.168.192.140
(private/internal).<br>• MITRE techniques T1110.001 and T1021.004 are associated with the alert. |
197 | **Reasonable interpretation** | • The pattern of repeated failures suggests a password-guessing or brute-force attempt.<br>• The
internal IP could belong to an authorized administrator, a scheduled scan, or a compromised workstation. |
198 | **Important uncertainty** | • No successful login or subsequent activity has been recorded.<br>• The legitimacy of the source IP and
the intent behind the failures remain unknown. |
199
200 ## Investigation Steps
201
202 1. **Review SSH authentication logs** on the target host for any successful logins from 192.168.192.140 or related timestamps.
203 2. **Cross-reference the source IP** against the organization's asset inventory and known administrative devices.
204 3. **Check for scheduled scans or automated tools** that might target SSH on internal hosts, and verify their credentials.
205 4. **Inspect the originating host** (192.168.192.140) for signs of compromise: unusual processes, outbound connections, or recent
configuration changes.

```

```

208 ## Recommended Remediation
209
210 - **If the source is unauthorized,** consider blocking or rate-limiting SSH access from 192.168.192.140 pending confirmation.
211 - **Enforce MFA and strong password policies** for all SSH accounts to mitigate future brute-force attempts.
212 - **Implement account lockout thresholds** to automatically suspend accounts after a defined number of failed attempts.
213 - **Restrict SSH access** to known administrative IP ranges and enforce key-based authentication where possible.
214 - **Monitor SSH logs continuously** for repeated failure patterns and integrate alerts with a SIEM for real-time correlation.
215
216 ## SOC Analyst Conclusion
217
218 **Classification:** Suspicious
219 **Rationale:** The alert confirms repeated SSH authentication failures from an internal IP, but there is no evidence of successful login,
credential compromise, or lateral movement. The source could be legitimate or malicious; further evidence is required to confirm
compromise.
220
221 **Evidence is insufficient to confirm compromise.**
222
223 ---
224
225 # 📄 Report Information
226
227 - Generated by: **AI SOC Assistant**
228 - Report Format: **Markdown (.md)**
229 - Analysis Engine: **OpenRouter LLM**
230 - Severity Source: **Deterministic Risk Engine**
231

```

Figure 13,14: Generated SOC incident report

11.7 Overall Result

The testing demonstrated that the finalized v4.8 AI-SOC Assistant can process a structured Wazuh alert through the complete SOC analysis pipeline. The system successfully performed alert formatting, applicable threat-intelligence processing, deterministic risk assessment, incident classification, evidence assessment, AI-assisted analysis, and incident report generation.

12. LIMITATIONS AND FUTURE IMPROVEMENTS

12.1 Limitations

Although the AI-SOC Assistant provides a structured SOC analysis pipeline, the current implementation has several limitations.

First, the project processes structured Wazuh alert data and does not directly function as a complete enterprise SIEM deployment. The current testing therefore focuses on the alert-analysis pipeline rather than continuous production monitoring.

Second, threat intelligence availability depends on the applicable external service. Private internal IP addresses are not submitted to public IP reputation services such as AbuseIPDB, so their reputation cannot be determined through that source.

Third, the deterministic risk engine uses predefined scoring rules. Although this provides consistent and explainable results, the scoring model does not dynamically learn from historical incidents.

Fourth, AI-generated analysis depends on the quality and completeness of the evidence supplied to the AI component. The system therefore explicitly instructs the AI not to treat assumptions or MITRE technique associations as proof of successful malicious activity.

Finally, the current project is designed as a personal cybersecurity/SOC analysis project and has not been validated as a production enterprise SOC platform.

12.2 Future Improvements

Future versions could expand the project with additional capabilities such as integration with live Wazuh alert streams, broader threat-intelligence sources, historical incident correlation, analyst feedback mechanisms, automated case management, and more advanced visualization.

These improvements are outside the scope of the finalized **v4.8** implementation. Version 4.8 was intentionally frozen as the completed project architecture, and additional feature modules are not part of the current implementation

13. CONCLUSION

The AI-SOC Assistant v4.8 project demonstrates the development of an evidence-based cybersecurity analysis pipeline designed to support SOC analyst activities. The system processes Wazuh security alerts through a structured sequence of alert parsing, alert formatting, applicable threat intelligence, deterministic risk assessment, incident classification, evidence assessment, AI-assisted analysis, and incident report generation.

A key design principle of the project is the separation between deterministic security analysis and AI-assisted reasoning. The Risk Engine provides a consistent and explainable risk assessment, while the AI component uses the structured evidence and context to produce a human-readable SOC analysis. This approach helps reduce unsupported conclusions and maintains a clear distinction between confirmed evidence, suspicious observations, and unconfirmed possibilities.

Testing with the SSH brute-force sample alert demonstrated that the finalized system can successfully process an alert and produce a structured SOC assessment. The tested alert received a risk score of **55/100**, a **HIGH** risk level, and a **SUSPICIOUS** incident status. The system also classified the incident as **SSH Brute Force** and generated a corresponding incident report.

Overall, the project achieved its defined objectives for version **v4.8** and provides a practical foundation for understanding SOC alert analysis, cybersecurity evidence assessment, deterministic risk scoring, threat intelligence integration, MITRE ATT&CK context, and AI-assisted security operations.

Version 4.8 represents the finalized and frozen implementation of the AI-SOC Assistant within the defined scope of this personal project.

14. REFERENCES

14.1 Cybersecurity and SOC References

1. [MITRE ATT&CK](#) — MITRE ATT&CK knowledge base for adversary tactics, techniques, and mitigations.
2. [Wazuh Documentation](#) — Wazuh documentation for security monitoring, alerts, rules, and detection capabilities.
3. [AbuseIPDB](#) — IP reputation and abuse-reporting service used for applicable public IP threat-intelligence lookups.

14.2 Python References

4. [Python Documentation](#) — Python language and standard-library documentation.
5. [Requests Documentation](#) — Documentation for the Python HTTP library used for API communication.

14.3 AI / LLM Reference

6. [OpenRouter Documentation](#) — Documentation for the LLM API interface used by the AI analysis component.

15. APPENDICES

Appendix A — Project Structure

AI-SOC-Assistant/

```
|  
|  
|─ app/  
|   |─ abuseipdb.py  
|   |─ ai.py  
|   |─ alert_formatter.py  
|   |─ config.py  
|   |─ context_builder.py  
|   |─ detector.py  
|   |─ evidence.py  
|   |─ incident_classifier.py  
|   |─ ip_utils.py  
|   |─ main.py  
|   |─ mitre.py  
|   |─ output.py  
|   |─ parser.py  
|   |─ prompts.py  
|   |─ report.py  
|   |─ risk_engine.py  
|   |─ threat_rating.py  
|   |─ ti_formatter.py
```

```
|   └─ ti_manager.py
|
|─ docs/
|─ reports/
|─ screenshots/
|─ sample_alerts/
|   └─ ssh_bruteforce.json
|
|─ .env
|─ .gitignore
|─ LICENSE
|─ README.md
└─ requirements.txt
```

Appendix B — Sample Wazuh Alert

```
sample_alerts > {} ssh_bruteforce.json > {} mitre
1  {
2    "rule": {
3      "id": "5763",
4      "level": 10,
5      "description": "Multiple SSH authentication failures"
6    },
7    "agent": {
8      "name": "ubuntu-agent"
9    },
10   "srcip": "192.168.192.140",
11   "timestamp": "2026-08-04T19:30:15Z",
12   "mitre": {
13     "id": [
14       "T1110.001",
15       "T1021.004"
16     ],
17     "tactic": [
18       "Credential Access",
19       "Lateral Movement"
20     ]
21   }
22 }
```

Figure 10: Sample Wazuh SSH brute-force alert used for system testing

Appendix C — Console Execution Output

===== WAZUH ALERT =====

Rule ID: 5763

Rule Level: 10

Description: Multiple SSH authentication failures

Agent: ubuntu-agent

Source IP: 192.168.192.140

Timestamp: 2026-08-04T19:30:15Z

MITRE IDs: T1110.001, T1021.004

MITRE Tactics: Credential Access, Lateral Movement

Cybersecurity Knowledge

Technique: Password Guessing

Tactic: Credential Access

Description: Attackers attempt to guess passwords for valid user accounts.

Mitigation: Use MFA, account lockout policies, and strong passwords.

Technique: SSH

Tactic: Lateral Movement

Description: Attackers use SSH to move laterally or remotely access systems.

Mitigation: Restrict SSH access, use key authentication, and monitor login attempts.

===== RISK ASSESSMENT =====

Risk Score : 55/100

Risk Level : HIGH

Incident Status : SUSPICIOUS

Confidence : MEDIUM

Risk Factors:

- Wazuh Rule Level 10+: +30
- Private Internal IP: +5
- AbuseIPDB Score 0%: +0
- MITRE T1110.001 Password Guessing: +10
- MITRE T1021.004 SSH: +10

===== INCIDENT CLASSIFICATION =====

Incident Type : SSH Brute Force
Attack Category : Credential Attack
Primary Technique : T1110.001 - Password Guessing
Protocol : SSH

===== EVIDENCE ASSESSMENT =====

Confirmed Evidence:

- Wazuh rule 5763 generated the alert.
- Rule description: Multiple SSH authentication failures.
- Rule level: 10.
- Source IP observed: 192.168.192.140.
- MITRE techniques associated with the alert: T1110.001, T1021.004.

Suspicious Observations:

- The observed authentication failures may indicate password-guessing activity.
- The activity requires investigation to determine whether the source is authorized or potentially compromised.

Unconfirmed:

- Successful authentication has not been confirmed.
- Credential compromise has not been confirmed.
- Lateral movement has not been confirmed.
- Command execution has not been confirmed.
- Host compromise has not been confirmed.

===== AI SOC ANALYSIS =====

Executive Summary

A Wazuh rule (5763) flagged multiple failed SSH logins originating from the internal address ****192.168.192.140****. The alert is classified as a credential-guessing attempt (MITRE T1110.001) and potentially a lateral-movement probe (T1021.004). No evidence of successful authentication, credential compromise, or subsequent activity has been observed. The incident is therefore ****suspicious****; further investigation is required to determine whether the source is an authorized user, a legitimate security scan, or an attacker-controlled host.

Attack Assessment

Category	Observation
Confirmed behavior	Wazuh generated a Level 10 alert for multiple SSH authentication failures. Source IP is 192.168.192.140 (private/internal). MITRE techniques T1110.001 and T1021.004 are associated with the alert.
Reasonable interpretation	The pattern of repeated failures suggests a password-guessing or brute-force attempt. The internal IP could belong to an authorized administrator, a scheduled scan, or a compromised workstation.
Important uncertainty	No successful login or subsequent activity has been recorded. The legitimacy of the source IP and the intent behind the failures remain unknown.

Investigation Steps

1. ****Review SSH authentication logs**** on the target host for any successful logins from 192.168.192.140 or related timestamps.
2. ****Cross-reference the source IP**** against the organization's asset inventory and known administrative devices.
3. ****Check for scheduled scans or automated tools**** that might target SSH on internal hosts, and verify their credentials.
4. ****Inspect the originating host**** (192.168.192.140) for signs of compromise: unusual processes, outbound connections, or recent configuration changes.
5. ****Validate account lockout and MFA settings**** for the accounts targeted by the failures to assess potential impact.

Recommended Remediation

- ****If the source is unauthorized,**** consider blocking or rate-limiting SSH access from 192.168.192.140 pending confirmation.
- ****Enforce MFA and strong password policies**** for all SSH accounts to mitigate future brute-force attempts.
- ****Implement account lockout thresholds**** to automatically suspend accounts after a defined number of failed attempts.
- ****Restrict SSH access**** to known administrative IP ranges and enforce key-based authentication where possible.
- ****Monitor SSH logs continuously**** for repeated failure patterns and integrate alerts with a SIEM for real-time correlation.

SOC Analyst Conclusion

****Classification:**** Suspicious

****Rationale:**** The alert confirms repeated SSH authentication failures from an internal IP, but there is no evidence of successful login, credential compromise, or lateral movement. The source could be legitimate or malicious; further evidence is required to confirm compromise.

****Evidence is insufficient to confirm compromise.****

Figure 11: Complete end-to-end execution of the AI-SOC Assistant v4.8

Appendix D — Generated Incident Report

```
reports > Incident_Report_20260817_222223.md > abc # Evidence Disclaimer
1  # AI SOC INCIDENT REPORT
2
3  **Incident ID:** IR-20260817-222223
4
5  **Generated:** 17 August 2026, 10:22:23 PM
6
7  **Severity:** HIGH
8
9  **Risk Score:** 55/100
10
11 **Incident Status:** SUSPICIOUS
12
13 **Confidence:** MEDIUM
14
15 ---
16
17 # Incident Context
18
19 ===== INCIDENT CONTEXT =====
20
21 ### WAZUH ALERT
22
23 Rule ID: 5763
24 Rule Level: 10
25 Description: Multiple SSH authentication failures
26
27 Agent: ubuntu-agent
28
29 Source IP: 192.168.192.140
```

```
31  Timestamp: 2026-08-04T19:30:15Z
32
33  MITRE IDs: T1110.001, T1021.004
34
35  MITRE Tactics: Credential Access, Lateral Movement
36
37  Cybersecurity Knowledge
38
39  Technique: Password Guessing
40  Tactic: Credential Access
41  Description: Attackers attempt to guess passwords for valid user accounts.
42  Mitigation: Use MFA, account lockout policies, and strong passwords.
43
44  Technique: SSH
45  Tactic: Lateral Movement
46  Description: Attackers use SSH to move laterally or remotely access systems.
47  Mitigation: Restrict SSH access, use key authentication, and monitor login attempts.
48
49  =====
50
51  ### THREAT INTELLIGENCE
52
53  ===== Threat Intelligence =====
54
55  IP Address : 192.168.192.140
56
57  IP Type : Private Internal Address
```



```
58
59 Threat Intelligence : Not Applicable
60
61 Reason :
62 Private IP addresses are not indexed by public
63 Threat Intelligence platforms.
64
65 Threat Rating : ● INTERNAL
66
67 Lookup Status : SKIPPED
68
69 =====
70
71 ### DETERMINISTIC RISK ASSESSMENT
72
73 Risk Score:
74 55/100
75
76 Risk Level:
77 HIGH
78
79 Incident Status:
80 SUSPICIOUS
81
82 Confidence:
83 MEDIUM
84
85 =====
```

```

87  ### INCIDENT CLASSIFICATION
88
89  Incident Type:
90  SSH Brute Force
91
92  Attack Category:
93  Credential Attack
94
95  Primary Technique:
96  T1110.001 - Password Guessing
97
98  Protocol:
99  SSH
100
101  =====
102
103  ### CONFIRMED EVIDENCE
104
105  - Wazuh rule 5763 generated the alert.
106  - Rule description: Multiple SSH authentication failures.
107  - Rule level: 10.
108  - Source IP observed: 192.168.192.140.
109  - MITRE techniques associated with the alert: T1110.001, T1021.004.
110
111  =====
112
113  ### SUSPICIOUS OBSERVATIONS
114
115  - The observed authentication failures may indicate password-guessing activity.
116  - The activity requires investigation to determine whether the source is authorized or potentially compromised.
117
118  =====
119
120  ### UNCONFIRMED
121
122  - Successful authentication has not been confirmed.
123  - Credential compromise has not been confirmed.
124  - Lateral movement has not been confirmed.
125  - Command execution has not been confirmed.
126  - Host compromise has not been confirmed.
127
128  =====
129
130  ### SOC ANALYSIS REQUIREMENTS
131
132  Analyze this incident as a professional SOC analyst.
133
134  IMPORTANT:
135
136  1. Clearly separate confirmed facts from reasonable interpretations
137  |   and unconfirmed possibilities.
138
139  2. Do not claim compromise unless the supplied evidence confirms it.
140

```

```
140
141 3. Do not claim successful authentication unless successful
142 | authentication is explicitly present in the evidence.
143
144 4. Do not claim lateral movement unless there is direct evidence
145 | supporting lateral movement.
146
147 5. Do not treat the absence of threat intelligence as proof that
148 | an IP address is safe.
149
150 6. Consider threat intelligence when assessing the incident.
151
152 7. If threat intelligence conflicts with the observed Wazuh behavior,
153 | explicitly describe the conflict and consider both sources.
154
155 8. Respect the deterministic risk assessment and incident classification,
156 | but do not treat either one as proof of compromise.
157
158 9. Treat MITRE ATT&CK mappings as technique classifications,
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163
164 11. Recommend investigation steps based only on available evidence.
165
166 12. Do not invent usernames, processes, malware, commands,
167 | successful logins, network activity, or other events.
168
169 13. For private/internal IP addresses, consider legitimate administrative
170 | activity, authorized security scanning, testing, or a potentially
171 | compromised internal host as possible explanations.
172
173 14. Do not overstate confidence.
174
175 15. If the available evidence is insufficient to establish compromise,
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177
178 | "Evidence is insufficient to confirm compromise."
179
180 16. The final conclusion must be based on supplied evidence,
181 | not assumptions or external information.
182
183 =====
```

```

186
187 # 🛡️ AI SOC Analysis
188
189 ## Executive Summary
190 A Wazuh rule (5763) flagged multiple failed SSH logins originating from the internal address 192.168.192.140. The alert is classified
as a credential-guessing attempt (MITRE T1110.001) and potentially a lateral-movement probe (T1021.004). No evidence of successful
authentication, credential compromise, or subsequent activity has been observed. The incident is therefore suspicious; further
investigation is required to determine whether the source is an authorized user, a legitimate security scan, or an attacker-controlled
host.
191
192 ## Attack Assessment
193
194 | Category | Observation |
195 |-----|-----|
196 | Confirmed behavior | • Wazuh generated a Level 10 alert for multiple SSH authentication failures.<br>• Source IP is 192.168.192.140
(private/internal).<br>• MITRE techniques T1110.001 and T1021.004 are associated with the alert. |
197 | Reasonable interpretation | • The pattern of repeated failures suggests a password-guessing or brute-force attempt.<br>• The
internal IP could belong to an authorized administrator, a scheduled scan, or a compromised workstation. |
198 | Important uncertainty | • No successful login or subsequent activity has been recorded.<br>• The legitimacy of the source IP and
the intent behind the failures remain unknown. |
199
200 ## Investigation Steps
201
202 1. Review SSH authentication logs on the target host for any successful logins from 192.168.192.140 or related timestamps.
203 2. Cross-reference the source IP against the organization's asset inventory and known administrative devices.
204 3. Check for scheduled scans or automated tools that might target SSH on internal hosts, and verify their credentials.
205 4. Inspect the originating host (192.168.192.140) for signs of compromise: unusual processes, outbound connections, or recent
configuration changes.
206
207 ## Recommended Remediation
208
209 - If the source is unauthorized, consider blocking or rate-limiting SSH access from 192.168.192.140 pending confirmation.
210 - Enforce MFA and strong password policies for all SSH accounts to mitigate future brute-force attempts.
211 - Implement account lockout thresholds to automatically suspend accounts after a defined number of failed attempts.
212 - Restrict SSH access to known administrative IP ranges and enforce key-based authentication where possible.
213 - Monitor SSH logs continuously for repeated failure patterns and integrate alerts with a SIEM for real-time correlation.
214
215 ## SOC Analyst Conclusion
216
217 Classification: Suspicious
218 Rationale: The alert confirms repeated SSH authentication failures from an internal IP, but there is no evidence of successful login,
credential compromise, or lateral movement. The source could be legitimate or malicious; further evidence is required to confirm
compromise.
219
220 Evidence is insufficient to confirm compromise.
221
222 ---
223
224 # 📄 Report Information
225
226 - Generated by: AI SOC Assistant
227 - Report Format: Markdown (.md)
228 - Analysis Engine: OpenRouter LLM
229 - Severity Source: Deterministic Risk Engine
230
231

```

Figure 12: Generated SOC incident report

Appendix E — Git Versioning

The screenshot displays the GitHub interface for the 'AI-SOC-Assistant' repository. The top bar shows the repository name, a 'Public' badge, and interaction buttons for Pin, Watch (0), Fork (0), and Star (0). Below the bar, the 'main' branch is selected, with 1 branch and 3 tags. A search bar and 'Add file' button are present. The file structure on the left lists: app (v4.8 - Fix alert evidence description handling, 5 days ago), sample_alerts (v4.7 - Refine evidence-based SOC analysis and reporting, last week), .gitignore (Add AI-powered incident report generation, 2 weeks ago), LICENSE (Initial commit, 2 weeks ago), README.md (Initial commit, 2 weeks ago), docs (Create docs, 2 weeks ago), report (Create report, 2 weeks ago), requirements.txt (Add Gemini AI configuration and prompt module, 2 weeks ago), and screenshots (Create screenshots, 2 weeks ago). The right sidebar contains 'About' (No description, website, or topics provided), 'Releases' (3 tags, Create a new release), and 'Packages'.

Commits on Aug 9, 2026

- v4.4 integrate risk assessment into AI SOC context** (huzafa-wk committed last week) 15ada58
- v4.3 Add deterministic risk assessment to incident pipeline** (huzafa-wk committed last week) 3758ae9
- v4.2 Evidence-Based AI Reasoning** (huzafa-wk committed last week) d7a368e

Commits on Aug 7, 2026

- v4.1 - Improved Threat Intelligence handling** (huzafa-wk committed 2 weeks ago) 88fe94b
- v4.0 - Complete AI SOC pipeline with Threat Intelligence integration** (huzafa-wk committed 2 weeks ago) b0bc63b

Commits on Aug 13, 2026

- v4.8 - Fix alert evidence description handling** (huzafa-wk committed 5 days ago) 65e4d05

Commits on Aug 12, 2026

- v4.7 - Refine evidence-based SOC analysis and reporting** (huzafa-wk committed last week) 41f3d8b
- v4.6(new) - complete evidence-based SOC alert pipeline** (huzafa-wk committed last week) d3d4383

Commits on Aug 11, 2026

- v4.6 - Add deterministic risk and evidence assessment** (huzafa-wk committed last week) 0a71ff9
- v4.5 - Add deterministic incident classification** (huzafa-wk committed last week) 65ba359

Figure 13: Git versioning and finalized v4.8 project state